Hearing Heritage: Imaginary and Immersive Soundscapes

Alexander, Cate Cleo
cate.alexander@mail.utoronto.ca
University of Toronto, Canada

Knight, Lauren
lauren.knight@mail.utoronto.ca
University of Toronto, Canada

In the Titanic Belfast Museum, the sounds of a creaking ship echo through the exhibit. Snippets of song trace the path through the Museum of Broadway, from Oklahoma to the Phantom of the Opera. Heady, pounding music reverberates through the Museum of Sex. While museums have historically served as 'empires of sight' (Stewart 1999), specialized museums increasingly incorporate audio technologies into their exhibits. Through examining the use of sonic technologies in museum spaces, we argue that these technologies increase accessibility, dismantle colonial 'empires of sight,' and broaden the possibilities of accessing cultural heritage through other sensory modalities. We investigate these technologies through ethnographic reflections and field recordings in selected museums as well as artistic autoethnographic experiments (Elliot / Culhane 2016) with AI sound generation using Imaginary Soundscape. Through this work, we connect AI to broader histories of sonic innovation/intervention in museums.

We suggest that attending to the role of audio in museums fosters unique possibilities for affective experiential processing (Ahmed 2014; Massumi 1995). Early forms of audio boxes through to newer pre-recorded audio guides indicate a lineage of sonic media interventions within these spaces. More recently, museums have integrated sonic media broadly for larger collective listening experiences — as background material to foster an acoustic sensorial relationship to the visual material, as oral storytelling, and as immersive, affective experiences in otherwise dimly lit rooms. Referencing the historical evolution of sound in museums through to current technologies and future techno imaginaries (AI), we will conduct ethnographic and artistic ethnographic research on the use of sound in a selection of universal and specialized museums to examine their possibilities and limitations as immersion and affective experimental/experiential processing. In sum, we ask: In what ways does the use of sound embolden immersive and affective relationships to cultural heritage in museums? How does the addition of sonic media in these spaces subvert or challenge the dominance of visual material in museums?

We situate our research within soundscape studies and the possibilities of sound for fostering immersive environments (Feld 2001; Schafer 1977; Thompson 2002). Schafer argues that while we can choose to visually separate from a space, the sonic environment is all-encompassing: "We have no ear lids. We are condemned to listen" (Schafer 1986). For Schafer, we are always sonically immersed. Helmreich (2008) complicates this understanding of immersion by pointing to the significance of mediating technologies. We bring this focus on media and technology to our study of museums, drawing upon previous studies of museum soundscapes (Schoer 2014; Smart 2024) and Salmouka & Gazi's (2021) classification of sound in museums as "informative, interpretive, immersive."

We further situate our research within affect theory to address the possibilities and impacts of sound within sites of cultural heritage and particularly museums. Alison Landsberg argues that today's culturally dominant forms of engagement with history are experiential, and that people attempt to bring the past closer in affective, material, and tactile ways (2015). This "quest for history as a psychic need" (Cvekovitch 2003: 268) is reflected in museum spaces that use sonic technologies to make the past feel tangible, exciting, and immersive for visitors. While audio guides — the most visible of sonic interventions in museums — have a long history of use for increasing accessibility, there is significant debate over the balance between emotion and neutrality (Hutchinson / Eardley 2020; Felix / Fleming 2023). We connect these debates over emotion and neutrality to sound studies which argues for the inherent affectual impact of sound, and our ethnographic observations of intentional sound design to elicit emotional responses/experiences.

In the first phase of our research, we will conduct ethnographic field work in four specialized museums: the New York Transit Museum (opened 1976), the Museum of Sex (opened 2002), Titanic Belfast (opened 2012), and the Broadway Museum opened 2022). These museums were chosen for their unique use of sound design, as described in the introduction. We will conduct site visits to collect ethnographic reflections (field notes and field recordings) detailing the role of sound in developing immersive, accessible, and affective relationships to the museums and their visual material. We will situate these observations within

broader histories and scholarship of sonic intervention/innovation within museums. In the second phase of our research, we will use AI sound generation to explore two universal museums — The Brooklyn Museum (opened 1897) and The Royal Ontario Museum (opened 1914) — as sites of artistic ethnographic experimentation. Our artistic experimentation will involve capturing 20-30 pictures within the museums and uploading this material to an AI soundscape generator: Imaginary Soundscapes. A web based installation, Imaginary Soundscapes allows visitors to virtually walk within Google Street View or upload images to

immerse "themselves into imaginary soundscapes generated with deep learning models" (Imaginary Soundscapes n.d.). The platform utilizes two deep-learning models to analyze an image and translate this data into an audio spectrogram. Pulling on a repository of approximately 48,000 – 55,000 sound files (at time of writing) sourced from Freesound, the algorithm retrieves one file that is determined to align most closely with the visual scene provided. This audio file is then looped to develop an imaginary soundscape for the visual image. We will analyze the AI generated audio for our images through ethnographic reflection and listening practice (Schafer 1977). Through coding the characteristics of the soundscapes, we will also work to reverse engineer the logics of the AI sound generator. This artistic ethnographic experimentation will allow us to reflect on the relationship between visuality and sonic media within museums, while simultaneously engaging in an analysis of the possibilities, limitations, and ethics of AI generated material or technologies within cultural heritage sites.

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